

Loop complexity, java

1) static long single (n)

c=0

```
for (int i=0 ; i<n ; i++)
```

loop run upto n

c++ even if c++ increase

it run upto n

time complexity $O(n)$

n

c

1000

1000

2000

2000

3000

3000

2) static long nested (int n)

c=0

```
for (int i=0 ; i<n ; i++) {
```

```
    for (int j=0 ; j<n ; j++) {
```

c++

return c

$O(n) \times O(n)$

c

n

i

j

0

5

0

0

1

5

1

1

$O(n^2)$



3) static long triangle (n)

c=0

for (int i=0; i<n; i++) {

for (int j=0; j<i; j++) {

c++

n=5

i	j	
0	0	0 ✓
1	1	1 ✓
2	2	2 1
3	3	3 2

$O(n(n-1)/2)$

$O(n^2)$

4) halving

c=0

for (int i=n; i>1; i/=2)

c++

n=5

i=5

5>1 ✓

c=1

i = 5/2 = 2

2>1 ✓

i = 2/2 = 1

1>1 ✗

c=2

$O(\log n)$

iterative division used 5/2 = 2 not

2.5



ADSUL'S TECHNICAL CAMPUS



5) static tag $n \log n(n)$
 $c=0$

```
for (int i=0; i<n; i++)
    for (int j=1; j<n; j+=2)
        c++
```

	i	j	
$n=5$	0	1	
$i=0$	0	2	$2 < 5$
$0 < 5$		4	$4 < 5$ ✓
for $i=0$ inner loop run 1		8	at 3rd time it
for $i=1$ inner loop again run		8	$8 < 5$ ✗

So inner loop executed for 3 times only
 inner loop for 5

so outer loop run 5 times

but

when

$i=2$

$2 < 5$ ✓

time
 $5 \times 3 = 15$
 $n(\log n) = \checkmark$

6) sequential (n)

c=0

for (int i=0 ; i<n ; i++)

c++

for (int i=0 ; i<n ; i++)

c++

n=5

first for loop run for 5 time

2nd for loop run for 5 time

$O(n) + O(n)$

$O(2n)$ ignore constant

$O(n)$

7) SqrtLoop (n)

c=0

for (long i=1 ; i*i<n ; i++)

c++

n=5

c	i	
0	1	$1 \times 1 < 5$ ✓
1	2	$2 \times 2 < 5$ ✓
2	3	$3 \times 3 < 5$ ✗

$(O(\sqrt{n}))$



8) Constant

$c=0$

for (int i=0; i<100; i++)

++

Here for loop executed at up to constant value
100

So complexity of constant is $O(1)$